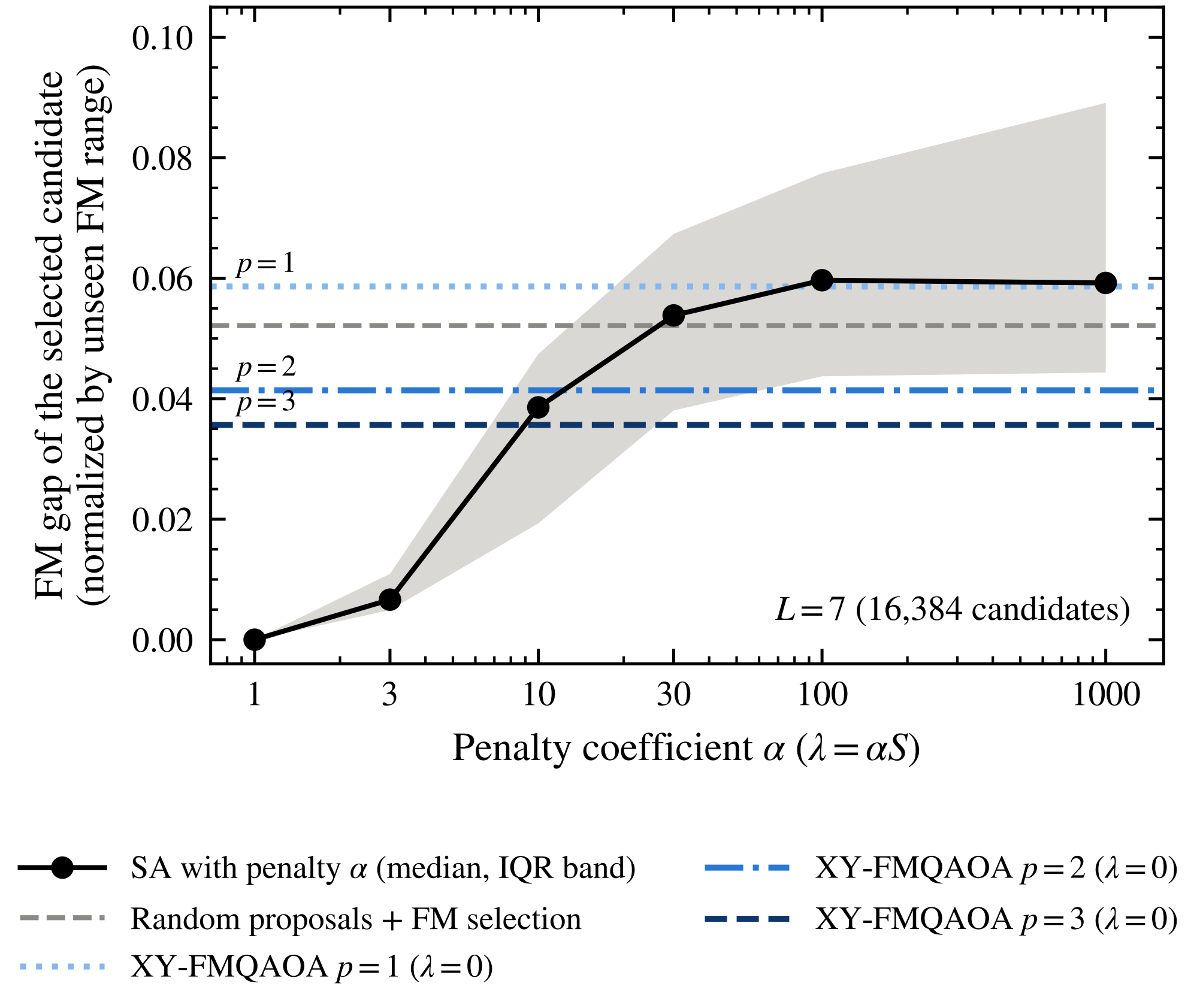
FM gap of the selected candidate
(normalized by unseen FM range)
0.10
0.08
0.06
0.04
0.02
0.00
$p=1$
$p=2$
$p=3$
$L=7$ (16,384 candidates)
1
3
10
30
100
1000
Penalty coefficient α ($\lambda=\alpha S$)
SA with penalty α (median, IQR band)
Random proposals + FM selection
XY-FMQAOA $p=1$ ($\lambda=0$)
XY-FMQAOA $p=2$ ($\lambda=0$)
XY-FMQAOA $p=3$ ($\lambda=0$)